Improving target-disease association prediction through a graph neural network with credibility information

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Abstract

Identifying effective target-disease associations (TDAs) can alleviate the tremendous cost incurred by clinical failures of drug development. We develop CreaTDA, an end-to-end deep learning-based framework that effectively learns latent feature representations of targets and diseases to facilitate TDA prediction. We also propose a novel way of encoding credibility information obtained from literature to enhance the performance of TDA prediction and predict more novel TDAs with real evidence support from previous studies. Compared with state-of-the-art baseline methods, CreaTDA achieves substantially better prediction performance on both the whole TDA network and its sparse sub-networks containing the proteins associated with few known diseases. Our results demonstrate that CreaTDA can provide a powerful and helpful tool for identifying novel target-disease associations, thereby facilitating drug discovery.

Methods

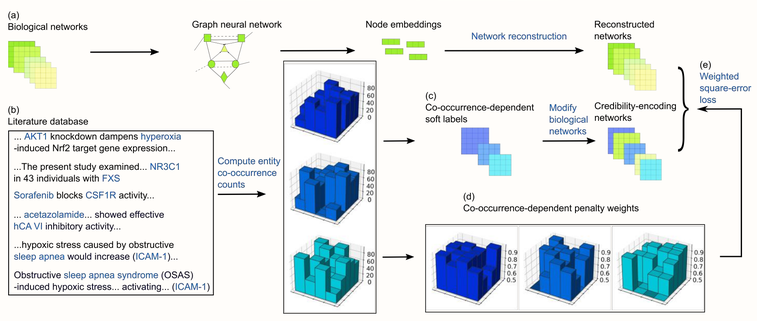

Figure 1: Overview of the CreaTDA pipeline. **(a)** Obtaining node embeddings from biological networks. **(b)** Computing entity co-occurrence counts in the PubMed database. **(c)** Co-occurrence-dependent soft labels. **(d)** Co-occurrence-dependent penalty weights. **(e)** Reconstructing the credibility-encoding networks containing the soft labels through minimizing a weighted square-error loss derived based on the penalty weights.

The name "CreaTDA" stands for **Cr**edibility-**e**ncoding gr**a**ph neural network for **TDA** prediction. As shown in Figure 1, CreaTDA computes node embeddings from biological networks. The entity co-occurrence counts in the PubMed database are then encoded as credibility information in the labels and loss-weighting scheme during optimization.

Results

We performed two five-fold cross-validation tests on a target-disease association (TDA) network derived from the Comparative Toxicogenomics Database [1]. The data was split by individual entries of the TDA adjacency matrix ("entry-wise" cross-validation) or by *clusters* of columns of the adjacency matrix ("cluster-wise" cross-validation). We compared the area under the receiver operating characteristic curve (AUROC) and the area under the precision-recall curve (AUPR) of CreaTDA with four baseline models that achieved state-of-the-art performance in heterogeneous graph prediction tasks: GTN [2], RGCN [3], HGT [4], and DTINet [5]. CreaTDA obtained overall superior cross-validation results (Table 1).

Table 1: Cross-validation results. The mean ± SD of the AUROC and AUPR scores are shown. The results where CreaTDA outperformed all baseline methods are presented in boldface.

	GTN	RGCN	HGT	DTINet	CreaTDA
Entry-wise cross-validation					
AUROC	0.953 ± 0.002	0.974 ± 0.001	0.950 ± 0.002	0.859± 2e-5	**0.986 ± 2e-4**
AUPR	0.822 ± 0.017	0.915 ± 0.004	0.846 ± 0.006	0.658± 1e-5	**0.967 ± 5e-4**
Cluster-wise cross-validation					
AUROC	0.725 ± 0.003	0.738 ± 0.014	0.569 ± 0.012	0.815 ± 0.007	0.814 ± 0.007
AUPR	0.397 ± 0.004	0.332 ± 0.013	0.211 ± 0.006	0.503 ± 0.018	**0.516 ± 0.016**

We gauged the credibility of the predicted novel TDAs, whose credibility information was *not* encoded during training, by examining their co-occurrence counts (C_r values) in the PubMed database. Compared with the baseline and control models, CreaTDA predicted more novel TDAs with high C_r values and yielded a higher correlation between the top output values and their corresponding C_r values (Fig 2), indicating that the novel TDAs predicted by CreaTDA were more likely to be biologically relevant.

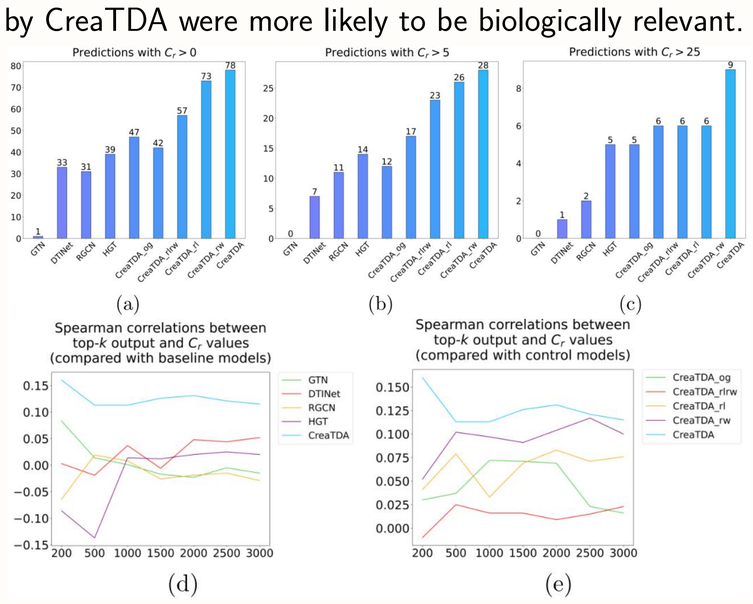

Figure 2: Examining the credibility of model predictions. **(a)**-**(c)** document the numbers of predictions among the top-200 novel predictions with $C_r > 0, 5, 25$, respectively. **(d)** and **(e)** plot the Spearman correlations between the output values of the top-k ($k = 200, 500, 1000, 1500, 2000, 2500, 3000$) predictions and their corresponding C_r values, with **(d)** comparing CreaTDA with the baseline models and **(e)** comparing CreaTDA with the four control models developed in our ablation study.

We examined the robustness of CreaTDA to the effect of exposure bias, a common phenomenon in recommendation systems where the unobserved interactions are often misrepresented as negative preferences [6]. In our TDA prediction task, the 0-labeled TDAs are not necessarily "negative" associations. Due to exposure bias, the models generally produce fewer meaningful TDA predictions for those proteins/diseases with fewer known TDAs and often have difficulty learning their latent feature representations.

As shown in Fig 3, compared with all baseline models, CreaTDA yields a lower correlation between the row-wise maximum values over the 0-labeled entries of the reconstructed TDA matrix and the row-wise sums of the original TDA matrix (i.e., the number of diseases associated with the corresponding protein), indicating that the predicted values of TDAs depend less on the amount of known information. CreaTDA also achieves superior performance on the sparse sub-networks of the original TDA network. These results suggest that CreaTDA exhibits robustness against the negative effect of exposure bias.

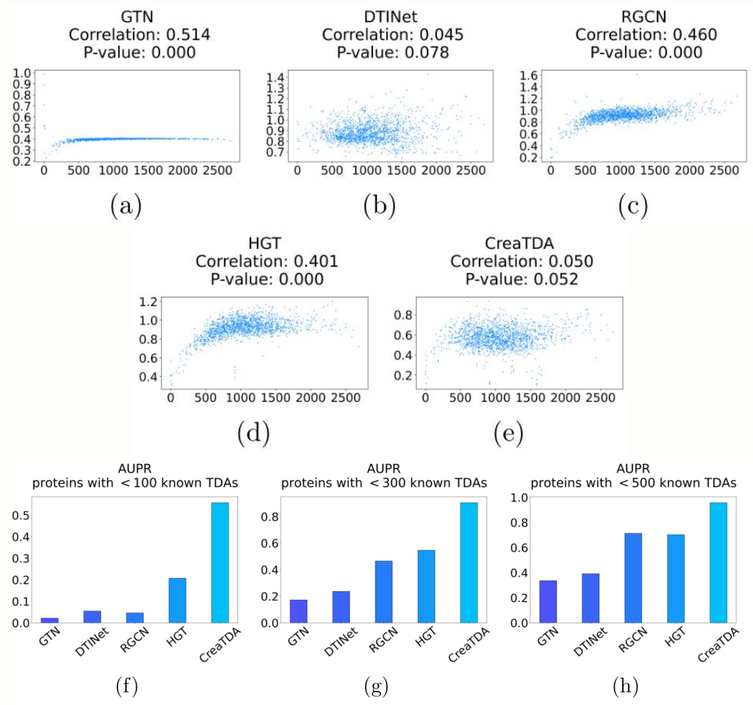

Figure 3: Examining the robustness against the effect of exposure bias. **(a)**-**(e)** plot the row-wise maximum values over the 0-labeled entries of the reconstructed TDA matrix (y-axis) against the row-wise sums of the original TDA matrix (x-axis) for the baseline models and CreaTDA. The Spearman correlations between these two vectors and their P-values are also reported. **(f)**-**(h)** present the AUPR scores on the sparse sub-networks of the whole TDA network containing proteins associated with few known TDAs.

Finally, we validated the top-200 predicted novel TDAs by searching for literature evidence, revealing potential disease targets. For instance, the epidermal growth factor receptor is a potential target for respiratory syncytial virus diseases, and the vascular endothelial growth factor-A may be a novel target for virus diseases in general.

Conclusion

In summary, by encoding credibility information into a graph neural network pipeline, the CreaTDA framework enjoys a superior prediction performance, predicts novel targets that are more likely to be biologically relevant, and is more robust to the effect of exposure bias. All these results suggest CreaTDA can provide a powerful and helpful tool to advance the drug discovery process.

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